

DEPARTMENT OF MATHEMATICS  
UNIVERSITY OF MARYLAND  
GRADUATE WRITTEN EXAM

August 2024

ALGEBRA (Ph.D. Version)

**Instructions to the student**

- a. Answer all six questions; each will be assigned a grade from 0 to 10.
  - b. Use a different booklet for each question. Write the problem number and your **code number** (**not** your name) on the outside of the booklet.
  - c. Keep scratch work on separate pages in the same booklet.
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1. (a) Let  $G$  be a group, let  $H$  be a finite normal subgroup of  $G$ , and let  $P$  be a Sylow  $p$ -subgroup of  $H$ . Show that if  $P$  is normal in  $H$  then  $P$  is normal in  $G$ .  
(b) Give an example of a group  $G$ , a normal subgroup  $H$  of  $G$ , and a normal subgroup  $K$  of  $H$  such that  $K$  is not normal in  $G$ .
2. (a) Let  $M$  be an  $n \times n$  matrix with entries in a field  $F$  and suppose the minimal polynomial  $m(X)$  of  $M$  has degree  $n$ , so  $m(X) = X^n + a_{n-1}X^{n-1} + \cdots + a_1X + a_0 \in F[X]$ . Show that  $M$  is similar over  $F$  to

$$\begin{pmatrix} 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & -a_1 \\ 0 & 1 & \cdots & 0 & -a_2 \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & 1 & -a_{n-1} \end{pmatrix}$$

(you may not quote results about existence of rational canonical form; that's what you're proving).

- (b) Let  $A$  and  $B$  be  $3 \times 3$  matrices with real entries such that  $A$  and  $B$  are not the identity matrix  $I$ . Suppose  $A^3 = I$  and  $B^3 = I$ . Show that  $A$  and  $B$  are similar over the reals.
3. Suppose that  $R$  is a non-zero commutative ring with 1. Let  $N$  the set of nilpotent elements of  $R$  (that is, elements  $x$  such that some power  $x^n = 0$ ). You may use, without proof, the facts that  $N$  is an ideal and equals the intersection of all prime ideals of  $R$ . Prove the equivalence of the following three statements:
  - (a) Every nonunit of  $R$  is nilpotent.
  - (b)  $R/N$  is a field.
  - (c)  $R$  has exactly one prime ideal.
4. Let  $R$  be a commutative ring with 1 and let  $M$  be an  $R$ -module. We say that  $M$  is flat if, whenever  $A \rightarrow B$  is an injective map of  $R$ -modules, the map  $A \otimes_R M \rightarrow B \otimes_R M$  is injective.

Show that if  $R$  is a PID and  $M$  is finitely generated, then  $M$  is flat if and only if  $M$  is torsion-free.

5. Let  $L/K$  be a finite Galois extension of fields with Galois group  $G$ . Suppose that whenever  $f(X) \in K[X]$  is an irreducible polynomial with a root in  $L$ , the splitting field  $F$  of  $f(X)$  over  $K$  satisfies  $[F : K] = \deg f(X)$ . Show that every subgroup of  $G$  is normal.
6. Suppose the group  $G$  has the following character table ( $A, B, C, D, E$  are the conjugacy classes in  $G$ , and  $\chi_0, \dots, \chi_5$  are the irreducible characters of  $G$ ):

|          | $A$ | $B$ | $C$ | $D$ | $E$ |
|----------|-----|-----|-----|-----|-----|
| $\chi_0$ | 1   | 1   | 1   | 1   | 1   |
| $\chi_1$ | 1   | -1  | 1   | 1   | -1  |
| $\chi_2$ | 2   | 0   | 2   | -1  | 0   |
| $\chi_3$ | 3   | 1   | -1  | 0   | -1  |
| $\chi_4$ | 3   | -1  | -1  | 0   | 1   |

The class of the identity element is  $A$  and the numbers of elements in some of the conjugacy classes are  $|A| = 1, |B| = 6, |C| = 3, |D| = 8$ .

- (a) How many elements are in  $E$ ? Justify your answer.
- (b) Show that  $G$  has a normal subgroup  $K$  with  $[G : K] = 2$ .
- (c) Let  $f(g) = \chi_3^3$  (it is the character of  $\rho_3 \otimes \rho_3 \otimes \rho_3$ , but you don't need to prove this fact). Decompose  $f$  as a sum of irreducible characters.